CHAPTER IV
PRESENTATION ANALYSIS AND INTERPRETATION OF DATA

This chapter presents the respondent characteristics, monitoring needs and expected care support most relevant to ElderCareAI. Elderly and caregiver results are reported separately. Replace the bracketed fields with verified results.

Report these items whether results support or challenge the system. Retain the complete questionnaire and remaining results in the appendix.

Profile of the respondents

The profile focuses on elderly age, health conditions and wearable comfort, plus caregiver workload, app familiarity and monitoring difficulties.

Age of Elderly Respondents

Form A — elderly respondents. Usable group responses: [N]. Valid answers to this item: [n_valid].

INSERT VERIFIED BAR CHART
Age distribution of elderly respondents
Use the categories and actual counts below

Figure 1: **Age distribution of elderly respondents**

Table 1

Age distribution of elderly respondents

Category	Frequency	Percentage
60-69 years old	[n]	[%]
70–79 years old	[n]	[%]
80 years old and above	[n]	[%]
Total valid answers	[n_valid]	[%]

Percentage = category frequency ÷ valid respondents answering this item × 100. Show the actual group denominator and any missing responses; start the bar-chart frequency axis at zero.

Figure 1 shows that [category] was reported by [n] respondents ([%]), compared with [n] ([%]) for [category]. [Describe the largest category and any ties after verifying the counts.] This describes the older adults represented and provides context for readable screens and assisted use. Age alone does not establish digital ability.

Reported Health Conditions

Form A — elderly respondents. Usable group responses: [N]. Valid answers to this item: [n_valid].

INSERT VERIFIED BAR CHART
Reported health conditions of elderly respondents
Use the categories and actual counts below

Figure 2: **Reported health conditions of elderly respondents**

Table 2

Reported health conditions of elderly respondents

Category	Frequency	Percentage
Heart condition	[n]	[%]
Diabetes	[n]	[%]
Hypertension (High Blood Pressure)	[n]	[%]
Respiratory condition	[n]	[%]
Other (specified)	[n]	[%]

Percentage = category frequency ÷ valid respondents answering this item × 100. Show the actual group denominator and any missing responses; start the bar-chart frequency axis at zero.

Multiple answers allowed: percentages can total more than 100%. Count respondents selecting each option, not the total number of selections. Preserve or code specific Other answers.

The form has no explicit None option. Missing or unselected health-condition answers do not prove absence of a condition.

Figure 2 shows that [category] was reported by [n] respondents ([%]), compared with [n] ([%]) for [category]. [Describe the largest category and any ties after verifying the counts.] This provides care context for health monitoring. Conditions are self-reported; they are not diagnoses established by the survey or app.

Comfort with Wearing a Tracker

Form A — elderly respondents. Usable group responses: [N]. Valid answers to this item: [n_valid].

INSERT VERIFIED BAR CHART
Elderly respondents' comfort with wearing a tracker
Use the categories and actual counts below

Figure 3: **Elderly respondents' comfort with wearing a tracker**

Table 3

Elderly respondents' comfort with wearing a tracker

Category	Frequency	Percentage
Very Comfortable	[n]	[%]
Comfortable	[n]	[%]
Neutral	[n]	[%]
Uncomfortable	[n]	[%]
Very Uncomfortable	[n]	[%]
Total valid answers	[n_valid]	[%]

Percentage = category frequency ÷ valid respondents answering this item × 100. Show the actual group denominator and any missing responses; start the bar-chart frequency axis at zero.

Figure 3 shows that [category] was reported by [n] respondents ([%]), compared with [n] ([%]) for [category]. [Describe the largest category and any ties after verifying the counts.] This describes the practical acceptability of wearing Galaxy Watch6. Report discomfort as clearly as comfort.

Number of Elderly Patients Under Care

Form B — caregivers. Usable group responses: [N]. Valid answers to this item: [n_valid].

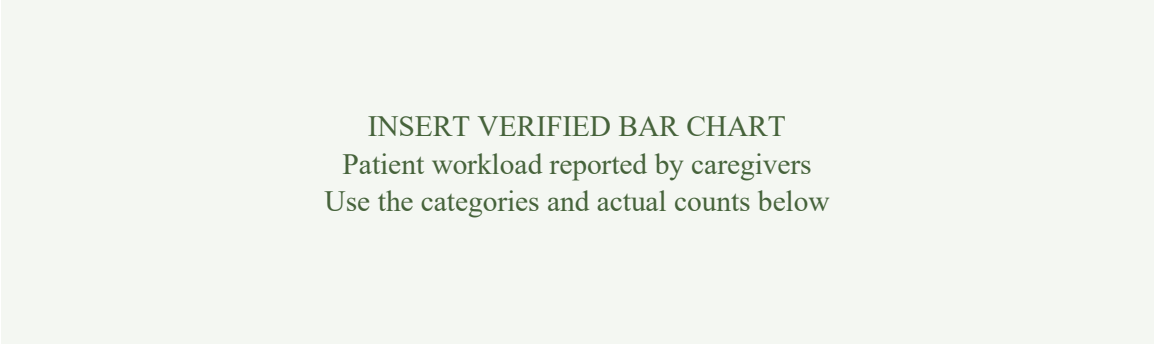


Figure 4: **Patient workload reported by caregivers**

Table 4

Patient workload reported by caregivers

Category	Frequency	Percentage
1–5 patients	[n]	[%]
6–10 patients	[n]	[%]
11–20 patients	[n]	[%]
More than 20 patients	[n]	[%]
Total valid answers	[n_valid]	[%]

Percentage = category frequency ÷ valid respondents answering this item × 100. Show the actual group denominator and any missing responses; start the bar-chart frequency axis at zero.

Figure 4 shows that [category] was reported by [n] respondents ([%]), compared with [n] ([%]) for [category]. [Describe the largest category and any ties after verifying the counts.] Workload provides context for reviewing patient records and receiving alerts. It does not prove that the system reduces workload.

Familiarity with Work Related Mobile Applications

Form B — caregivers. Usable group responses: [N]. Valid answers to this item: [n_valid].

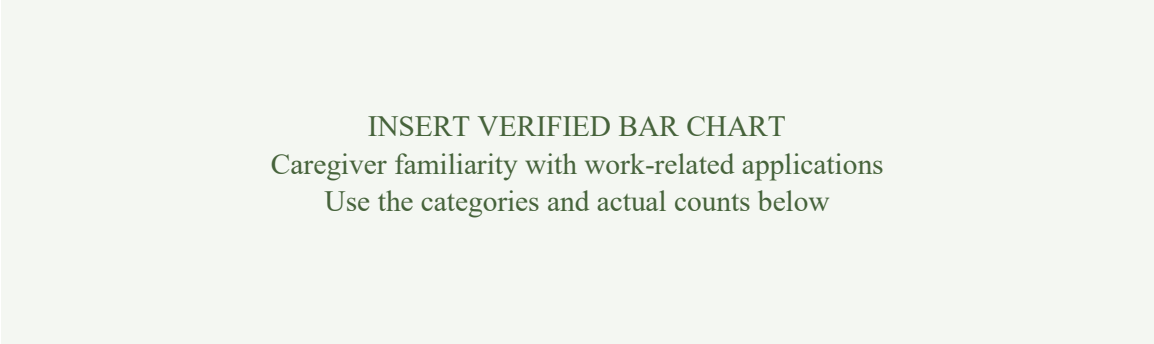


Figure 5: **Caregiver familiarity with work-related applications**

Table 5

Caregiver familiarity with work-related applications

Category	Frequency	Percentage
Very Familiar	[n]	[%]
Familiar	[n]	[%]
Unfamiliar	[n]	[%]
Very Unfamiliar	[n]	[%]
Total valid answers	[n_valid]	[%]

Percentage = category frequency ÷ valid respondents answering this item × 100. Show the actual group denominator and any missing responses; start the bar-chart frequency axis at zero.

Figure 5 shows that [category] was reported by [n] respondents ([%]), compared with [n] ([%]) for [category]. [Describe the largest category and any ties after verifying the counts.] This helps contextualize onboarding and interface accessibility. Preserve the four response options; this question has no Neutral category.

Health Indicators Challenging to Monitor Manually

Form B — caregivers. Usable group responses: [N]. Valid answers to this item: [n_valid].

INSERT VERIFIED BAR CHART
Health indicators caregivers find challenging to monitor
Use the categories and actual counts below

Figure 6: **Health indicators caregivers find challenging to monitor**

Table 6

Health indicators caregivers find challenging to monitor

Category	Frequency	Percentage
Heart Rate	[n]	[%]
Blood Oxygen Level (SpO2)	[n]	[%]
Sleep Pattern	[n]	[%]
Stress Level	[n]	[%]
Skin Temperature	[n]	[%]
Physical Activity / Fitness	[n]	[%]
Other (specified)	[n]	[%]

Percentage = category frequency ÷ valid respondents answering this item × 100. Show the actual group denominator and any missing responses; start the bar-chart frequency axis at zero.

Multiple answers allowed: percentages can total more than 100%. Count respondents selecting each option, not the total number of selections. Preserve or code specific Other answers.

Figure 6 shows that [category] was reported by [n] respondents ([%]), compared with [n] ([%]) for [category]. [Describe the largest category and any ties after verifying the counts.] This identifies reported monitoring difficulties relevant to the vital cards. It does not establish clinical accuracy or prove that every metric is continuously available.

Key findings on the existing monitoring process

These selected baseline items describe monitoring coverage and burden. They include positive and negative statements, so the discussion can identify both current strengths and difficulties. A denotes elderly respondents and B denotes caregivers; their results must not be pooled.

SA = Strongly Agree (5); A = Agree (4); N = Neutral (3); D = Disagree (2); SD = Strongly Disagree (1). Each response column contains a count. M = item mean.

Table 7

Selected existing process ratings

Item	Statement	SA	A	N	D	SD	M
A-B1	My health vitals (e.g., heart rate, blood pressure, temperature) are regularly checked by my caregiver.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
A-B5	I experience delays in receiving medical attention when I need it urgently. [negative]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
B-B1	I am able to effectively monitor the health vitals of all elderly patients under my care.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
B-B2	I face difficulties in tracking the health status of multiple elderly patients simultaneously. [negative]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
B-B5	Manually monitoring the health of elderly patients is time-consuming. [negative]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

Table 7 indicates that elderly respondents reported [finding about regular checks and delays]. Caregivers reported [finding about monitoring coverage, simultaneous tracking or time burden]. These findings describe the current care setting and provide context for ElderCareAI; they are not evidence of improvement after deployment.

Mean = (5×SA + 4×A + 3×N + 2×D + SD) ÷ valid answers to that item. Report missing answers separately. Higher agreement with [negative] items indicates a greater reported problem. Do not calculate a combined satisfaction mean from these mixed-direction items.

A-B1 mentions blood pressure in the existing care process. ElderCareAI does not currently measure blood pressure. Retain the original question wording when presenting survey results.

Acceptance and expected support for care tasks

Present willingness, concerns and expected usefulness as perceptions. These selected items support discussion of wearable adoption, Elle assistance and a consolidated view of available health information.

Table 8

Selected acceptance and expected usefulness ratings

Item	Statement	SA	A	N	D	SD	M
A-C1	I would be willing to wear a smartwatch that continuously monitors my health.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
A-C5	I am concerned about wearing a health-monitoring device throughout the day. [negative]	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
B-C5	An AI chatbot that provides guidance and support to caregivers will be a helpful tool in daily caregiving tasks.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
B-C7	Monitoring of heart rate, blood oxygen, sleep, stress, skin temperature, and fitness activity will give a more complete picture of a patient's health.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

A-C5 is a concern about wearing a device all day. Report it separately from willingness; do not describe agreement with that concern as positive acceptance.

Table 9

Selected expected benefits of key features

Item	Statement	SA	A	N	D	SD	M
B-D2	The sleep tracking feature will give me valuable insights into patients' rest quality and help me adjust care plans accordingly.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
B-D10	Automated caregiver notifications when abnormal vitals are detected will significantly improve my emergency response time.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]
B-D12	The automated health report generation feature will reduce the time I spend on manual documentation of patient health data.	[ ]	[ ]	[ ]	[ ]	[ ]	[ ]

Tables 8 and 9 show [supported findings about willingness and concerns], while caregivers expected [supported usefulness findings]. Report any neutral or unfavorable responses. These opinions concern proposed benefits; they do not demonstrate better sleep assessment, faster emergency response or less documentation time in actual use.

Use item means and five-category counts without an overall composite. Preserve the statements as administered. Only available recorded sleep and supported measurements can be shown by the app; the experimental stress estimate is not a clinical assessment.

Optional: if relevant comments were actually submitted, add one short paragraph summarizing recurring anonymous recommendations. Omit it when there are no comments; do not add invented themes.

Questionnaires: Form A elderly | Form B caregivers